INDEPENDENT BOUNDARY TRACE CONTROLLABILITY FOR A DISJOINT HEAT CASCADE

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Editorial note. Public 2109 is finite-dimensional nonlinear shape desensitization (Thm-Case1-Exact-FD); this independently controls arbitrary pairs of boundary fluxes and retains the original unique-continuation argument. It is not the direct zero-flux desensitization corollary.

Allowed standard prerequisites (editorial).

- Classical heat equation and functional analysis. Weak and transposition solutions, Sobolev trace maps and parabolic regularity, Holmgren unique continuation, Hilbert-space range/kernel duality, standard heat observability and convex minimization. The displayed Hadamard derivative formula is a standard supplied input. Target-specific constructions remain in the original proof.

Let $T > 0$ denote a horizon of time, ω and Θ be two open subsets of $\mathbb{R}^d$, $d \in \mathbb{N}^*$, and $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$.

For a bounded, connected and open set Ω of $\mathbb{R}^d$ of class $\mathscr{C}^2$, we consider the shape functional J_h, indexed by $h \in L^2(0,T;L^2(\omega))$, defined by

$$J_h(\Omega) = \frac{1}{2}\int_0^T \int_\Theta y_{\Omega,h}(t,x)^2\, dx\, dt, \tag{1}$$

where $y_{\Omega,h} \in L^2((0,T)\times\mathbb{R}^d)$ is defined on $(0,T)\times\Omega$ as the unique weak solution of

$$\left\{\begin{array}{rcll} \dfrac{\partial y}{\partial t} - \Delta y &=& \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega, \\ y &=& 0 & \text{on } (0,T)\times\partial\Omega, \\ y\,(0,\cdot) &=& 0 & \text{in } \Omega, \end{array}\right. \tag{2}$$

($\mathbb{1}_\omega$ denotes the characteristic function of the set ω) extended by 0 outside $(0,T)\times\Omega$.

In the sequel, given $\boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$, we introduce the family $\{\Omega_{\tau\boldsymbol{V}}\}_{\tau\in[0,\tau_{\boldsymbol{V}})}$ of domains defined by

$$\Omega_{\tau\boldsymbol{V}} = (\mathrm{Id} + \tau\boldsymbol{V})(\Omega_0).$$

Proposition 0.1 ([28, Proof of Prop. 1.1])**.** *Let $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$ and $h \in L^2(0,T;L^2(\omega))$. For all $\boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$, the mapping $\tau \mapsto J_h(\Omega_{\tau\boldsymbol{V}})$ is differentiable at $\tau = 0$ and*

$$\frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = \int_{\partial\Omega_0} \boldsymbol{V}\cdot\boldsymbol{n}\left(\int_0^T \partial_n y_0 \partial_n z_0\, dt\right) d\sigma, \tag{3}$$

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where the pair (y_0, z_0) solves the coupled forward-backward system

$$
(4)\qquad \begin{cases} \dfrac{\partial y_0}{\partial t} - \Delta y_0 &= \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y_0 &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y_0(0,\cdot) &= 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z_0}{\partial t} - \Delta z_0 &= \mathbb{1}_\Theta y_0 & \text{in } (0,T)\times\Omega_0,\\ z_0 &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z_0(T,\cdot) &= 0 & \text{in } \Omega_0. \end{cases}
$$

($\mathbb{1}_\Theta$ denotes the characteristic function of the set Θ.)

Definition 0.2. Let $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$. We say that the control function $h \in L^2(0,T;L^2(\omega))$ desensitizes J_h exactly at Ω_0 at the first order with respect to domain variations if

$$
(5)\qquad \forall\, \boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d), \quad \frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = 0.
$$

Let $\mathcal{E}$ be a linear subspace of $W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$. We say that the control function $h \in L^2(0,T;L^2(\omega))$ exactly desensitizes J_h for $\mathcal{E}$ at Ω_0 at the first order with respect to domain variations if

$$
(6)\qquad \forall\, \boldsymbol{V} \in \mathcal{E}, \quad \frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = 0.
$$

Given $\varepsilon > 0$, we say that the control function $h \in L^2(0,T;L^2(\omega))$ ε-approximately desensitizes J_h at Ω_0 at the first order with respect to domain variations if

$$
(7)\qquad \forall\, \boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d), \quad \left|\frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0}\right| \leq \varepsilon \|\boldsymbol{V}\|_{W^{3,\infty}(\mathbb{R}^d;\mathbb{R}^d)}.
$$

0.1. **Main results.** As we said, the aforementioned desensitizing problems will be strongly dependent on the relative geometry of the various sets ω, Θ and Ω_0, and in particular depending if the set $\omega \cap \Theta$ is empty or not, but we will always make the following minimal assumption on both sets ω and Θ:

($\mathrm{H}_{\omega,\Theta}$) ω and Θ are two nonempty open subsets of Ω_0.

Case $\omega \cap \Theta = \emptyset$ and $\Theta \Subset \Omega_0$. To be more precise, the first geometric setting we consider is the following:

$$
(8)\qquad \omega \text{ and } \Theta \text{ satisfy } (\mathrm{H}_{\omega,\Theta}), \quad \omega\cap\Theta = \emptyset, \quad \Theta \Subset \Omega_0, \text{ and } \Omega_0 \setminus \overline{\Theta} \text{ is connected.}
$$

Proposition 0.3. *Assume the geometric setting* (8)*. Then, given any $(f_1,f_2) \in L^2(0,T;L^2(\partial\Omega_0))^2$, for any $\varepsilon > 0$ and $\xi \in L^2(0,T;L^2(\Omega_0))$, there exists a control function $h \in L^2(0,T;L^2(\omega))$ such that the solution (y,z) of*

$$
(9)\qquad \begin{cases} \dfrac{\partial y}{\partial t} - \Delta y &= \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y(0,\cdot) &= 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z}{\partial t} - \Delta z &= \mathbb{1}_\Theta y & \text{in } (0,T)\times\Omega_0,\\ z &= 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z(T,\cdot) &= 0 & \text{in } \Omega_0, \end{cases}
$$

satisfies

$$
(10)\qquad \|\partial_n y - f_1\|_{L^2(0,T;L^2(\partial\Omega_0))} + \|\partial_n z - f_2\|_{L^2(0,T;L^2(\partial\Omega_0))} \leq \varepsilon.
$$

0.2. **Proof of Proposition 0.3.** Proposition 0.3 can be recast in an abstract form into the problem: show that

$$\overline{\operatorname{Ran}\mathcal{L}} = (L^2(0,T;L^2(\partial\Omega_0)))^2,$$

where $\mathcal{L}$ is the operator defined for $h \in L^2(0,T;L^2(\omega))$ by

$$\mathcal{L}h = (\partial_n y_h, \partial_n z_h) \in L^2((0,T)\times\partial\Omega_0)^2,$$

where (y_h, z_h) solves

$$\left\{\begin{array}{rcll} \dfrac{\partial y_h}{\partial t} - \Delta y_h &=& h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y_h &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y_h(0,\cdot) &=& 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z_h}{\partial t} - \Delta z_h &=& \mathbb{1}_\Theta y_h & \text{in } (0,T)\times\Omega_0,\\ z_h &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z_h(T,\cdot) &=& 0 & \text{in } \Omega_0. \end{array}\right. \tag{11}$$

Notice that $\mathcal{L}$ is bounded, thanks to [15, Th. 5, p. 382] and the continuity of the operator $f \in H^2(\Omega_0) \mapsto \partial_n f \in L^2((0,T)\times\partial\Omega_0)$. Therefore, by standard arguments from functional analysis, Proposition 0.3 is equivalent to showing the dual property $\operatorname{Ker}\mathcal{L}^* = \{0\}$ (we refer for instance to [7, Cor. 2.18]). Using the arguments developed in [28, Prop. 2.1], one can compute explicitly $\mathcal{L}^*$ and deduce that $\operatorname{Ker}\mathcal{L}^* = \{0\}$ is equivalent to the following unique continuation problem: If $(g_1, g_2) \in (L^2(0,T;L^2(\partial\Omega_0)))^2$, and (ψ, φ) solves

$$\left\{\begin{array}{rcll} -\dfrac{\partial \psi}{\partial t} - \Delta\psi &=& \mathbb{1}_\Theta\varphi & \text{in } (0,T)\times\Omega_0,\\ \psi &=& g_1 & \text{on } (0,T)\times\partial\Omega_0,\\ \psi(T,\cdot) &=& 0 & \text{in } \Omega_0,\\ \dfrac{\partial \varphi}{\partial t} - \Delta\varphi &=& 0 & \text{in } (0,T)\times\Omega_0,\\ \varphi &=& g_2 & \text{on } (0,T)\times\partial\Omega_0,\\ \varphi(0,\cdot) &=& 0 & \text{in } \Omega_0, \end{array}\right. \tag{12}$$

then, we have the following unique continuation property:

$$\psi = 0 \text{ in } (0,T)\times\omega \implies g_1 = g_2 = 0. \tag{13}$$

We now prove this unique continuation property. Let $(g_1, g_2) \in (L^2(0,T;L^2(\partial\Omega_0)))^2$ be such that the solution (ψ, φ) of (12) satisfies $\psi = 0$ in $(0,T)\times\omega$. From [27, (15.17–18), p. 86], we notably infer that $(\psi, \varphi) \in L^2((0,T)\times\Omega_0)$.

We first work in the set $(0,T)\times(\Omega_0 \setminus \overline{\Theta})$. There, ψ satisfies the usual backward heat equation

$$-\frac{\partial\psi}{\partial t} - \Delta\psi = 0 \quad \text{in } (0,T)\times(\Omega_0\setminus\overline{\Theta}),$$

where we do not specify any initial or boundary conditions. Therefore, since $\omega \subset \Omega_0 \setminus \overline{\Theta}$ and $\Omega_0 \setminus \overline{\Theta}$ is connected thanks to Assumption (8), using the Holmgren theorem for the heat equation (see e.g.[21, §8.6 & Th. 8.6.5]), we thus infer that $\psi = 0$ in $(0,T)\times(\Omega_0\setminus\overline{\Theta})$. In particular, using [34, Prop. 7.1.3], since ψ the unique weak solution of (12) in the sense of [34, Def. 7.1.2, p. 342], for any $\zeta \in H^2(\Omega_0)\cap H_0^1(\Omega_0)$ such that $\Delta\zeta \in H_0^1(\Omega_0)$, for any $t \in [0,T]$, we have

$$\int_0^t \int_{(0,T)\times\partial\Omega_0} g_1 \partial_n\zeta \, d\sigma dt = 0.$$

Differentiating this inequality with respect to t (which is possible because

$$g_1(\cdot,x)\partial_n\zeta(x) \in L^2(0,T) \quad \text{for almost all } x \in (0,T)\times\partial\Omega_0,$$

since the Neumann trace operator $f \in H^2(\Omega_0) \mapsto \partial_n f \in L^2(\partial\Omega_0)$ is well-defined), we obtain that for any $t \in [0,T]$, we have

$$\int_{\partial\Omega_0} g_1(t,\cdot)\partial_n\zeta d\sigma = 0.$$

Since $\{\zeta \in H^2(\Omega_0)\cap H_0^1(\Omega_0) \mid \Delta\zeta \in H_0^1(\Omega_0)\}$ is dense in $H^2(\Omega_0)\cap H_0^1(\Omega_0)$ for the H^2-norm, using the continuity of the Neumann trace operator

$$f \in H^2(\Omega_0) \longmapsto \partial_n f \in L^2((0,T)\times\partial\Omega_0),$$

we deduce that for any $\zeta \in H^2(\Omega_0)\cap H_0^1(\Omega_0)$, we have, for any $t\in[0,T]$,

$$\int_{\partial\Omega_0} g_1(t,\cdot)\partial_n\zeta d\sigma = 0.$$

Since the extended trace operator

$$f \in H^2(\Omega) \longmapsto (f_{|\partial\Omega_0}, \partial_n f_{|\partial\Omega_0}) \in H^{3/2}(\partial\Omega_0)\times H^{1/2}(\partial\Omega_0)$$

is surjective, taking into account that $0 \in H^{3/2}(\partial\Omega_0)$, we deduce that for any $h \in H^{1/2}(\partial\Omega_0)$, we have, for any $t\in[0,T]$,

$$\int_{\partial\Omega_0} g_1(t,\cdot) h d\sigma = 0.$$

Since $H^{1/2}(\partial\Omega_0)$ is dense in $L^2(\partial\Omega_0)$ for the $L^2(\partial\Omega_0)$-norm, we deduce that $g_1 = 0$ on $(0,T)\times\partial\Omega_0$, so that $\psi \equiv 0$ on $\Omega_0 \setminus \overline{\Theta}$ in the classical sense. Notably, $\partial_n\psi = 0$ on the whole boundary $\partial\Omega_0$ since $\Theta \Subset \Omega_0$.

We next remark that, by standard local regularity results for solutions of the heat equation, the function φ is smooth away from the boundary $\{0,T\}\times\partial\Omega_0$ (of class $\mathscr{C}^\infty((0,T)\times\Omega_0)$). Accordingly, $\mathbb{1}_\Theta\varphi$ belongs to $L^2(0,T;L^2(\Omega))$, and by standard maximal regularity results, the solution ψ belongs to

$$L^2(0,T;H^2_{\rm loc}(\Omega_0))\cap H^1(0,T;L^2_{\rm loc}(\Omega_0)).$$

In view of this regularity, and due to the fact that $\psi = 0$ in a neighbourhood of $(0,T)\times\partial\Omega_0$, we can multiply equation $(12)_{(1)}$ by φ and perform several integration by parts, using $g_1 = \partial_n\psi = 0$ on $\partial\Omega_0$, and $\psi = 0$ in $(0,T)\times(\Omega_0\setminus\overline{\Theta})$:

$$\begin{aligned}\int_0^T\int_\Theta |\varphi|^2\,dx\,dt &= \int_0^T\int_{\Omega_0}\varphi\Big(-\frac{\partial\psi}{\partial t} - \Delta\psi\Big)\,dx\,dt\\ &= -\int_{\Omega_0}\varphi(\cdot,x)\psi(\cdot,x)\,dx\Big|_0^T - \int_0^T\int_{\partial\Omega_0}\varphi\partial_n\psi d\sigma dt\\ &\qquad + \int_0^T\int_{\partial\Omega_0}\partial_n\varphi\psi d\sigma dt + \int_0^T\int_{\Omega_0}\Big(\frac{\partial\varphi}{\partial t} - \Delta\varphi\Big)\psi\,dx\,dt\\ &= 0.\end{aligned}$$

Therefore, $\varphi = 0$ in $(0,T)\times\Theta$, and by the classical unique continuation properties for the heat equation, $\varphi = 0$ in $(0,T)\times\Omega_0$, and in particular $g_2 = 0$. This concludes the proof of (13) for the solutions of (12), which proves that $\operatorname{Ker}\mathcal{L}^* = \{0\}$. Hence, Proposition 0.3 follows.

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